4.2 Preliminary Lift Planning

4.2.1 Two models can be used for lift traffic analysis and design:

a) The first model uses a calculation method based on mathematical formulae. This classical model uses the pure up-peak condition as this provides a well-defined, simple traffic pattern amenable to mathematical analysis. These calculations can be carried out manually, using a spreadsheet or by using computer based programme.

b) The second model is based on discrete digital simulation of the movement of lifts in a building and the passenger dynamics. This simulation model allows very complex situations to be analyzed and is generally capable of better evaluations. However, simulation approaches and algorithms vary with software ranging from crude to very sophisticated. With the varied approaches and lack of consensus on simulation approaches and how to apply simulation this standard does not define standards for simulation.

Generally, if a lift system (using a conventional landing call system) is sized correctly for the up-peak traffic pattern, all other traffic patterns should be adequately served. The notable exceptions to this, are:

1) Hotels where check-in/check-out clashes with meal times;
2) Hospitals;
3) Buildings that open at specified times (Example, stock exchanges, movie theatres, stadiums);
4) Buildings with 24×7 operation and shift changes (Example, BPO/ITES buildings);
5) Buildings with amenities located at higher floors (Example, roof top restaurant, cafeterias, health clubs);
6) Residential buildings (sometimes two-way traffic is considered);
7) Buildings with multiple entry levels;
8) Parking towers;
9) Schools where students change floors or buildings at the beginning and end of classes; and
10) Mixed-use buildings.

Methods of calculating the traffic handling capabilities of lifts were first devised for office buildings. In due course detailed modifications were devised to suit other applications without altering the basic principles. The application to office buildings is the most frequently used, and is outlined in **4.2.2** to **4.2.10** as a general approach to lifting.

Subsequently, traffic conditions other than up peak are examined as also building types other than office. It is recommended that calculations based on the classical model should be carried out in order to obtain an understanding of the various factors. A simulation may then be performed in order to cover special situations or to obtain information not provided by the calculation method.

4.2.2 *Population*

The first point to be ascertained from the owner/ developer or proposed occupier is the total building population. If a definite population figure is not available, then an estimation should be made using floor areas, more specifically, the net usable area (NUA). The NUA excludes circulation areas (stairs, corridors, foyers, etc), structural intrusions (columns, ducts, etc) and facilities (pantry, kitchens, toilets, conference rooms, training rooms, server rooms, storage areas, etc).

The number of people occupying the net usable area can vary according to the usage pattern or building type. Because of the criticality of establishing the population, it is recommended that the architect/engineer should prepare a typical test fit to establish an estimate of the population. The test fit estimates can be validated from population counts from buildings with similar usage patterns and types.

The designers need to understand the clear circulation and movement pattern of the occupants (and goods) of a building and design to facilitate the circulation and movement. Table 1 gives typical values for the density of population.

Table 1 Occupancy Area Per Person in Office Buildings
(*Clause* 4.2.2)

Sl No.	Building Type	Open Plan/ Hall Seating	Cellular Plan/ Cabins
(1)	(2)	(3)	(4)
i)	Premium	10 m^2	12 m^2
ii)	Regular	8 m^2	10 m^2
iii)	Low end	5 m^2	8 m^2

For certain specialized building functions such as BPO, areas ranging from 6 m^2 to 10 m^2 per person may be considered.

4.2.3 *Quantity of Service*

It is measured in terms of the total number of passengers handled during the peak 5 min period of the day. This is calculated by determining the number of trips made by the lifts over the peak 5 min period and then

multiplying it by the average number of passengers carried in each trip. The target handling capacity as a percentage of the building population is usually the estimated value of passenger arrival rate for the 5 min up-peak period. The minimum recommended quantity of service is given in Table 2.

This would vary depending on various factors like starting time (fixed or flexi-time), nearness to fast access routes like mass rapid transport systems, express ways, major roads, etc.

Table 2 Recommended Quantity of Service in Office Buildings
(*Clause* 4.2.3)

Sl No.	Type of Occupancy	Quantity of Service Percent
(1)	(2)	(3)
i)	Multi tenancy	10 to 15
ii)	Single tenancy	15 to 25

4.2.4 *Quality of Service*

It is very difficult to get real indications of passenger waiting time and lobby queuing from the classic calculation models. Therefore, interval is considered as the measure. During pure up-peak traffic, the interval or the average time between successive arrivals of the lift cars at the main lobby is generally considered as an indicator of passenger waiting time. Table 3 gives the quality of service based on interval.

If interval is used as an indicator of quality of service, it should be noted that passenger average waiting time depends on the car occupancy, the handling capacity *vis-a-vis* the passenger arrival rates, the control systems, etc. The average waiting time approximates 75 percent to 85 percent of the interval for average car occupancy of 80 percent of the actual lift capacity provided the handling capacity is equal to or better than the peak arrival rate. Beyond 80 percent car loading passenger average waiting time increases exponentially and cannot be approximated from interval. Table 4 gives the recommended quality of service measured as average waiting time.

Table 3 Recommended Quality of Service in Office Buildings (Based on Interval)
(*Clause* 4.2.4)

Sl No.	Quality of Service	Interval s
(1)	(2)	(3)
i)	Excellent	Less than 25
ii)	Very good	25 to less than 30
iii)	Good	30 to less than 35
iv)	Fair	35 to less than 40
v)	Poor	40 to less than 45
vi)	Unsatisfactory	45 and above

Table 4 Recommended Quality of Service in Office Buildings (Based on Average Waiting Time)
(*Clause* 4.2.4)

Sl No.	Class of Building	Average Waiting Time s
(1)	(2)	(3)
i)	Premium	< 25
ii)	Regular	25 to 35

4.2.5 *Traffic Analysis Equations*

The classical method to size a lift installation requires the determination of the time, in second, that it takes for a single lift to make a round trip around the building during the up-peak traffic condition. This is called the round trip time and is given by:

$$RTT = 2Ht_v + (S+1)(T - t_v) + 2Pt_p$$

where

N = number of floors above main terminal floors,

H = average highest reversal floor

$$= N - \sum_{i=1}^{N-1} (i/N)^P ,$$

S = average no of stops

$$= N\left[1 - \left(1 - \frac{1}{N}\right)^P\right],$$

P = average passengers carried

= 0.8 × maximum actual car capacity,

t_v = single floor transit time = average inter floor distance/rated speed,

T = door operating time (t_c+t_o) + single floor flight time $t_f(1)$, and

t_p = passenger transfer time.

The derivation of the round trip time equation assumes the following:

a) Traffic pattern corresponds to a pure up-peak.

b) Passengers arrive according to a rectangular probability distribution.

c) Lifts, on an average, fill to 80 percent of the actual car capacity by numbers irrespective of the weight of each passenger or the space that maybe occupied

d) All floors are equally populated.

e) Rated speed is achieved in a single floor jump.

f) Interfloor heights are equal.

g) Door dwell time does not exceed the calculated passenger transfer time.

Interval, $INT = RTT / L$, where L is the number of lifts in a single group.

$$\text{5 min handling capacity, } HC = \frac{300 \times P \times L \times 100 / \text{Population}}{RTT}$$

or

$$\frac{300 \times P \times 100 / \text{Population}}{INT}$$

The down peak round trip time is estimated as:

$$RTT_{\text{downpeak}} = Nt_{\text{v}} + (0.5S + 1)(T - t_{\text{v}}) + 2Pt_{\text{p}}$$

The mid-day round trip time is estimated as:

$$RTT_{\text{mid-day}} = 2Ht_{\text{v}} + 2S(T - t_{\text{v}}) + 4Pt_{\text{p}}$$

An estimation of interfloor performance can be obtained using:

$$AWT_{\text{interfloor}} = INT_{\text{uppeak}} + \left(0.22 + 1.78\frac{A_{\text{interfloor}}}{HC_{\text{uppeak}}}\right)$$

where $A_{\text{interfloor}}$ is the number of passengers arriving during the interfloor period.

4.2.6 *Configuration of Lifts: Number*

Determination of number of lifts depends on the required quality and quantity of service. It should be noted here that grouping of lifts and location should be given adequate attention.

If a bank of two lifts or more is required to meet the anticipated traffic requirements, they have to be oriented to facilitate grouping. Individual lifts will not achieve the required quantity and quality of service.

4.2.7 *Configuration of Lifts: Capacity*

Determination of car capacity depends on the required quality and quantity of service. A larger car, while improving the handling capacity might adversely impact the quality of service.

4.2.8 *Configuration of Lifts: Speed*

In addition to the speeds derived from the calculations for handling capacity and interval, the nominal travel time is used for selecting suitable rated speeds. The higher the building, the faster lifts are needed. The recommended values for nominal travel time in case of an office, commercial or hotel building are given in Table 5.

Table 5 Recommended Nominal Travel Time for Office, Commercial or Hotel Buildings
(*Clause* 4.2.8)

Sl No.	Level	Nominal Travel Time s
(1)	(2)	(3)
i)	Excellent	15 to 25
ii)	Good	> 25 to 35
iii)	Satisfactory	> 35 to 45

4.2.9 *Configuration of Lifts: Layout*

The shape and size of a passenger lift car bears a distinct relation to its efficiency in traffic handling. A study of the most suitable proportions for passenger lifts reveals that the width of the lift well entrance is the basic element in the determination of the best proportions. In other words, the width of the car is determined by the width of the entrance and the depth of the car is regulated by the loading per square metre permissible under this standard.

Centre opening doors are the most practicable and the most efficient entrance units for passenger lifts. Given the same door speed, the centre opening is much faster than the side opening type. For passenger lifts, wider cars are preferred as the ingress and egress efficiency is higher. For service/freight/ stretcher lifts deeper cars are preferred, so as to easily carry the stretcher/goods.

While sizing the hoist-ways, the possible construction inaccuracies have to be considered. Further where high speeds lifts are being considered the hoist-ways need to be designed to address the impact of the wind tunnel effect.

4.3 Planning for Specific Building Features

4.3.1 *Special Building Facilities*

Facilities like cafeteria, food courts, restaurants, gymnasia, etc, in the building impact the circulation patterns in a building and should be considered for lift traffic design.

4.3.2 *Basement Service*

Buildings are sometimes designed with car parks or other facilities at basement levels below the terminal floor. Such floors, if they are served by the main lifts have an adverse impact on the up-peak, down-peak and mid-day traffic patterns. During up-peak conditions, this may cause lifts to arrive at the main terminal already partly full causing confusion. In such a situation people have a tendency to push both up and down buttons resulting into deterioration of the lift service.

If basement floors are served by only a part of a lift group, passengers experience difficulty in selecting the correct lifts out of a group that will serve the basement. It is therefore better to provide basement service to the main lobby by a separate group of 'shuttle' lifts so as to avoid compromising the traffic handling capability of the main lift group.

4.3.3 *Multiple Entry Levels*

Some buildings have main entry points at more than one level. The effect of more than one main terminal is disruptive and adversely impacts effective circulation and movement. If there are more than one entrance

levels, means should be provided to bring all the routes to a single terminal floor. If this is not possible, then the lift system planning should take into account the extra times incurred stopping and loading at multiple entry floors. Another difficulty is in deciding whether the building population will use each entrance equally. In the absence of any guidance, the solution is to assume an entrance bias with an additional 10 percent and size the lifts to meet the additional required handling capacity.

The loss of lift efficiency because of multiple entry levels could be as high as 15 percent per additional entry level. Both basement service and multiple entry level buildings with lifts serving all floors also can impact the security of the building.

4.3.4 *Non-Smoking Buildings*

When buildings are defined as non-smoking buildings, designers need to factor in the additional load put on the lifts on account of the additional trips that smokers make to go outside the building. It is appropriate to provide pressurized smoking zones at the floor level rather than require smokers to make additional trips.

4.3.5 *Reserved Lifts*

When lifts have to be reserved for VIP personal movement, designers should not consider these lifts as part of the lift group.

4.3.6 *Zoning (Vertical)/Sky Lobbies in Very Tall Buildings*

As the number of floors served increases, the values of H and S also increase, adversely impacting the round trip time and performance of the lift system. This has led to the concept of zoning in tall buildings. In stacked zoning, a tall building is effectively divided into horizontal layers or stacks. RTT, quality and quantity of service can be arrived at separately for each zone. The desired level of service can be attained by adjusting two parameters in this case — the number of lifts and the number of floors in the zone. The round trip time for each zone is given by:

$$RTT = 2Ht_v + (S + 1)(T - t_v) + 2Pt_p + 2[t_{express} - t_f(1)]$$

where $t_{express}$ is the flight time from the main terminal to the express zone terminal/sky lobby.

The round trip time for shuttle lifts serving only 2 floors — main terminal and sky lobby is given by:

$$RTT = 2T + 2Pt_p$$

4.4 Planning For Specific Building Types

4.4.1 *Residential Buildings*

The peak traffic condition occurs during the morning down-peak when many adults and children are leaving for work and school at the same time. Population assumptions would also need to consider the service staff, drivers of cars, etc. For high rise buildings the delivery of newspapers and milk and disposal of garbage are also required to be considered.

In the luxury housing segment, where the number of ratio of service staff to residents is fairly high, separate service lifts may be required for door deliveries and service staff. Typical average daily population for a residential building is given in Table 6.

The passenger handling capacity shall be as given in Table 7. The intervals shall be as per values in Table 8. Table 9 gives the recommended quality of service

Table 6 Typical Average Population for Residential Buildings
(*Clause* 4.4.1)

Sl No.	Type of Apartment	Number of Bedrooms Per Apartment				
		1	2	3	3 With Servant Quarters	4 With or Without Servant Quarters
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	High end apartments:					
	a) Residents	2 to 3	3 to 4	5	5	5 to 6
	b) Resident service staff	0	0	0	1	1
	c) Floating service staff	1	2 to 3	3 to 4	3 to 4	4 to 5
ii)	Mid end apartments:					
	a) Residents	3	4	5	5	5 to 6
	b) Resident service staff	0	0	0	1	1
	c) Floating service staff	1	1	2 to 3	2 to 3	3 to 4
iii)	Low end apartments:					
	a) Residents	4 to 5	5 to 6	NA	NA	NA
	b) Floating service staff	0.5	0.5	NA	NA	NA

NOTE — Unless separate lifts are provided for service staff, due consideration for service staff shall be given while calculating the required number of lifts.

measured as average waiting time for residential buildings and Table 10 gives the nominal travel times.

Sizing of lifts should consider the requirement to shift the stretchers, heavy material, etc and it is recommended that at least one lift in each building should be a deep car which can accommodate a regular ambulance stretcher.

Table 7 Recommended Handling Capacity for Residential Buildings
(*Clause* 4.4.1)

Sl No.	Class of Building	Handling Capacity Percent
(1)	(2)	(3)
i)	High end building	> 8
ii)	Mid end building	6 to 8
iii)	Low end building	5 to 7

Table 8 Recommended Quality of Service for Residential Buildings (Based on Time Interval)
(*Clause* 4.4.1)

Sl No.	Class of Building	Interval s
(1)	(2)	(3)
i)	High end building	≤ 60
ii)	Mid end building	61 to 80
iii)	Low end building	81 to 100

Table 9 Recommended Quality of Service in Residential Buildings (Based on Average Waiting Time)
(*Clause* 4.4.1)

Sl No.	Class of Building	Average Waiting Time s
(1)	(2)	(3)
i)	High end building	≤ 30
ii)	Mid end building	31 to 45
iii)	Low end building	46 to 60

Table 10 Recommended Nominal Travel Time for Residential Buildings
(*Clause* 4.4.1)

Sl No.	Level	Nominal Travel Time s
(1)	(2)	(3)
i)	High end building	25 to 35
ii)	Mid end building	36 to 45
iii)	Low end building	46 to 60

NOTE — In case of super high rise buildings considering limit of lift speed as 10 m/s, the travel time may be longer than specified above.

4.4.2 *Hospitals*

Factors to be considered include number of staff and shift patterns, number of visitors and visiting hours, location of operation theatres, facilities, delivery of housekeeping supplies, waste disposal, evacuation procedures and segregation of sterile areas. When carrying out traffic analysis for hospitals, designers should consider establishing average car loading by volume rather than by weight. Arrival rates may approximate 10 percent to 15 percent and interval may be 30 s to 50 s.

The sizing of the lift car and doors should be such that a standard hospital stretcher and attendant can be easily accommodated. Where the hospital bed is likely to be moved the lift car should be able to accommodate the standard hospital bed including the auxiliary support equipment like oxygen cylinders, etc.

In large hospitals it is recommended that the lifts for the patients/hospital beds (or stretchers) should be separated from the lifts for staff/visitors.

NOTE — For hospital buildings, designers should also consider that in an emergency a number of patients would require to be evacuated on stretchers.

4.4.3 *Hotels*

The most demanding time is during the check-in and check-out period and two-way traffic occurs during this period with guests going to and from rooms and restaurants and in and out of the hotel. Calculations should be made assuming an equal number of up and down stops during this period. Average room occupancy may range from 1.5 to 2 persons and arrival rates from 10 to 15 percent. Security considerations need to be taken into account whilst establishing the circulation and lift requirements. Escalators should be employed for heavy short range movements such as from the lobby to banquet/ function level. Adequate number of service lifts need to be provided for service movement of housekeeping and room service staff as well as movement of material.

4.4.4 *Retail — Malls with Multiplexes*

Pedestrian movement in retail establishments like malls is generally centered on escalators, and lifts do not play a major part. However, provision should be made for movement of shopping trolleys, wheel chairs, perambulators and persons with limited mobility from one level to another. A commonly applied solution is the installation of inclined moving walks. Where lifts are provided, assumptions of lift car sizing should take into account space occupied by trolleys and shopping bags. Where multiplexes are located on the upper floors, the last movie of the day gets over after the mall is closed at which time the escalators might not be available. In such cases the lift provision has to be

adequate to handle the egress of the total multiplex population.

4.4.5 *Airports/Railway Stations*

While moving walks and escalators greatly improve the building circulation, the use of lifts has to be considered for persons with limited mobility and for movement of baggage trolleys from one level to another. The RTT equation of **4.2.5** may be used, but care may need to be taken in the assumptions of lift car occupancy levels taking into account space occupied by trolleys.

4.4.6 *Multi-Level Car Parking*

Multi-level car parks may be standalone public car parks or attached to office, retail or residential complexes. These car parks may be fully automated where drivers would leave the cars inside or on a trolley at the entry floor and the car is parked and retrieved automatically. The other alternative would be for the use of car lifts and/or ramps.

When the movement of the cars is dependent on car lifts, detailed study has to be carried out to establish the required number of car lifts ensuring that average car retrieval/parking time does not exceed 2 min. The sizing of the car lifts has to be adequate to fit the largest vehicle that is intended to be transported as well as adequate space to enable opening of the doors to enable evacuation of passengers in the eventuality of an entrapment. Designers will also need to take into account the probability of queues developing and provide for holding lanes.

4.4.7 *Multi-Level Car Parking Passenger Traffic*

When car lifts or ramps are the means for parking the cars, then means would need to be provided for the movement of passengers from and to the parking floors. If the main building lifts also serve the parking floors, it is to be noted that, performance will be adversely impacted due to multiple entry floors. Additional entry floors will also affect the security of the building. It is therefore recommended that separate lifts should be considered to move passengers. If separate parking lifts are provided, the basic RTT equation of **4.2.5** may be used. Average vehicle occupancy may be considered as 1.5 per car for office car parks, 4 for airports and retail and 2 elsewhere.

4.4.8 *Schools and Other Educational Institutions*

In schools and other educational institutions, the traffic flow would consist of peak demand for short duration that would exist just before the start or after finishing of a class or lecture. It is unlikely that an economical solution can be implemented for such high peak requirements. Therefore the design of the building has to be such that heavy stair usage is facilitated.

4.4.9 *Buildings with 24 × 7 Operations*

In buildings operating 24 × 7 (round the clock), the peak traffic conditions would not be a typical up peak but will occur during shift changes. The handling capacity should take into account the incoming as well as outgoing traffic. Calculations shall be made assuming both up and down stops during this period. Particular attention should be paid while designating the drop off points, these should be assigned to a level so as to not increase the burden on the system. Designers need to consider that the lifts in such buildings would have a significantly higher number of starts/stops per hour than a conventional building.

4.4.10 *Observatory*

The height at which the observatory is located and the number of people expected at the observatory level shall decide the speed and the number of lifts. The circulation logic is crucial in locating the observatory lifts. Also, if the observatory is connected with lounge/café/souvenir shops, etc, separate service lifts may need to be provided.

4.5 Other Considerations

Designers need to be aware that the door opening and closing times, acceleration and deceleration times, etc, would vary between the suppliers and equipment types and can impact the overall lift performance. The dispatch algorithms and controller responses would also vary between the suppliers and the equipment types and can impact the actual operational results.

Lifts will breakdown as well require to be shut-down for regular maintenance and repairs. Lift availability might be hampered during renovation of the building as a lift might be taken up for movement of material and debris. Designers should take into consideration the impact of such non-availability of lifts. The provision of well-located and easily accessible stairs can considerably lessen the demands on the lifts and therefore architect/engineer should consider this aspect in the layout.

4.5.1 *Lift Speed*

For passenger lifts in a building, the general recommendations as given in Table 11 may be followed.

4.5.2 *Quiet Operation of Lifts*

Every precaution should be taken with passenger lifts to ensure quiet operation of the lift doors and machinery. The insulating of the lift machine and any motor generator from the floor by rubber cushions or by a precast concrete slab with rubber cushions, prevents transmission of most of the noise. In this connection, see also good practice [8-5A(4)] and Part 8 'Building Services, Section 4 Acoustics, Sound Insulation and

Noise Control' of the Code for some useful recommendations.

Table 11 Lift Speed
(*Clause* 4.5.1)

Sl No. (1)	No. of Floors (2)	Speed m/s (3)
i)	Up to 6	0.6 to 1.0
ii)	7 to 15	1.0 to 1.5
iii)	16 to 20	1.5 to 1.75
iv)	21 to 30	1.75 to 2.5
v)	31 to 45	3.0 to 4.0
vi)	46 to 60	4.0 to 6.0
vii)	Above 60	6.0 and above

NOTES
1 Above table is considering buildings with average floor height of 3.0 m.
2 For office buildings average floor height is 4.2 m (*see* Table 5 and Table 10).
3 Finalizing the lift speed for any building depends on detailed traffic analysis because number of floors, area per floor, area per person and class of building are crucial factors, which serve as inputs for such traffic analysis.
4 Finalizing the speed for hotel depends on the number of keys per floor and similarly for an observatory depends on the number of expected footfall.
5 For any building, the speed of the lifts will have to be validated against the total population and travel height of the building.

5 LIFT ARRANGEMENTS AND PLANNING DIMENSIONS

The appropriate aspect of lift installation shall be discussed during the preliminary planning of the building with all concerned parties, namely, client, architect, consulting engineer and/or lift manufacturer. This enables the lift manufacturer to furnish the architect and/or consulting engineer with the proposed layout or *vice-versa*.

5.1 Exchange of Information

5.1.1 The guidelines laid down together with Fig. 2 will enable the preliminary scheme for the installation to be established. Figure 2 shows only some of the typical arrangements and variations are possible with respect to number of lifts and the layout of lifts especially when destination control system (DCS) is used instead of conventional group control system

Although the recommended outline for the various classes of lifts enable the general planning details to be determined by the architect/engineer, these should be finally settled at the earliest possible stage by detailed investigation with the purchaser's representative reaching agreement with the lift manufacturer (where necessary) before an order is finally placed. This will enable a check to be made and information to be exchanged on vital matters such as:

a) Number, capacity, speed and disposition of the lifts necessary to give adequate lift service in the proposed building;
b) Provision of adequate access to the machine room;
c) Loads which the lift will impose on the building structure, and the holes to be left in the machine room floor and cut-outs for wall boxes for push-buttons and signals;
d) Necessity for and type of insulation to minimize the transmission of vibration and noise to other parts of the building;
e) Special requirements of local authorities and other requirements set out in the 'planning permit';
f) Need for the builder to maintain accuracy of building as to dimensions and in-plumb;
g) Periods of time required for preparation and approval of relevant drawings for manufacturing and the installation of the lift equipment;
h) Requirements for fixing guide brackets to the building structure;
j) Time at which electric power will be required before completion to allow for testing;
k) Requirements for electrical supply feeders, etc;
m) Requirements for scaffolding in the lift well and protection of the lift well prior to and during installation of equipment; and
n) Delivery and storage of equipment.

5.1.2 *Information to be provided by Architect or Engineer*

As a result of preliminary discussions, the drawings of the building should give the following particulars and finished sizes:

a) Number, type and size of lifts and position of lift well;
b) Particulars of lift well enclosure;
c) Size, position, number and type of landing doors;
d) Number of floors served by the lift;
e) Height between floor levels;
f) Number of entrances;
g) Total headroom;
h) Provision of access to machine room;
j) Provision of ventilation and, if possible, natural lighting of machine room;
k) Height of machine room;
m) Depth of lift pit;
n) Position of lift machine, above or below lift well;